The University of the South Pacific

School of IT, Engineering, Mathematics & Physics

CS214: Design & Analysis of Algorithms

Assignment 1 – Semester II, 2025

Total Marks: 20%

Due Date: as shown on Moodle

This assignment covers both theoretical and practical aspects of this course. The marking rubrics is heavily based on *data & information management*. Later assignments will focus more on *abstraction & design*. Rubrics have been taken from ACS-SCIMS rubrics V1.0. You will investigate the impact of having different data structures and algorithms to solve a given task. Java is the only acceptable language for this assignment. For graphics, you may use Matlab. This assessment covers the following course learning outcomes:

CLO 1 – Evaluate the efficiency of algorithms.
CLO 2 – Assess the suitability of different algorithms for solving a given problem.

Download the Research Article Dataset given in Moodle sourced from Kaggle at https://www.kaggle.com/datasets/arashnic/urban-sound?resource=download. Read this data in your Java program by creating an Article class and storing it in a linked list and array list separately. Now do the following:

1. Pick any four known search algorithms from https://www.geeksforgeeks.org/searching-algorithms/ and enhance the code to make them generic for any user-defined types. Additionally, make every algorithm work for both data structures *linkedlist* and *arraylist*. You may have to create separate algorithms for each data structure.

   In total, you will have 8 algorithms (or 4 if generalized) altogether where all the algorithms should be able to search for an element in a list given data structure. [3%]

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Abstraction & Design	IX. No adequate analysis of algorithm for its efficiency and relevance. X. inappropriate design of algorithms (efficiency, structure and data structures) XI. plagiarism	VI. Appropriate choice of the architecture and design VII. Analysis of algorithm for its efficiency and relevance. VIII. Appropriate design of algorithms (efficiency, structure and data structures)	I. All satisfactory and II. Some form of creativity or innovation to architecture and design. III. Innovative or creative design of algorithms (efficiency, structure and data structures)	**1**	
Programming	I. Plagiarism II. Poor indentation, hard to read and follow the code III. Lots of bugs and/or errors IV. Program produces unexpected output V. No proper encapsulation, inheritance or polymorphism for object-oriented programming VII. Hard coding of data in the program. VIII. Poor structure	I. Able to write a complex code for a loosely defined problem IV. Appropriate encapsulation, inheritance or polymorphism for object-oriented programming. V. computer program produces expected output	I. All satisfactory and demonstrate very good programming skills.	**1**	
System Development/ Acquisition	XII. Majority or most preferred features do not work as expected	XII. Many required features work correctly	XII. All required features work correctly	**1**	

Sub Total & comments			

2. Race all the above algorithms against each other on random test cases using a reasonably sized array, i.e., they will execute simultaneously. Use a third-party graphics library for Java or MATLAB for easy visualization. You may also use MATLAB commands in Java. Refer to the YouTube tutorial: https://www.youtube.com/watch?v=bvhUWcHflLQ. [4%]

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Data and Information Management	VII. Very limited or undesirable data has been collected VIII. Data is not stored and managed using inappropriate tools. IX. Poor or incorrect inference is derived from the collected/given data	VII. Appropriate data has been collected VIII. data is store and managed using a standard tool IX. logical inference is derived from the collected/given data	I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	**1**	
System Development/ Acquisition	XII. Majority or most preferred features do not work as expected	XII. Many required features work correctly	XII. All required features work correctly	**3**	
Sub Total & comments					

3. Depending on the location of the key you may get different results. You must randomly pick the key in every run and in some cases pick a key that does not exist. So run each algorithm 30 times and collect the results (using counter values) to find the best, mean, and worst solutions. Which algorithm is fastest according to your empirical testing? [5%]

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Data and Information Management	VII. Very limited or undesirable data has been collected VIII. Data is not stored and managed using inappropriate tools. IX. Poor or incorrect inference is derived from the collected/given data	VII. Appropriate data has been collected VIII. data is store and managed using a standard tool IX. logical inference is derived from the collected/given data	I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	**3**	
System Development/ Acquisition	XII. Majority or most preferred features do not work as expected	XII. Many required features work correctly	XII. All required features work correctly	**2**	
Sub Total & comments					

4. Determine the worst-case time complexity of these algorithms with different data structures graphically. [5%]

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Data and Information Management	VII. Very limited or undesirable data has been collected VIII. Data is not stored and managed using inappropriate tools. IX. Poor or incorrect inference is derived from the collected/given data	VII. Appropriate data has been collected VIII. data is store and managed using a standard tool IX. logical inference is derived from the collected/given data	I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	**2**	
System Development/ Acquisition	XII. Majority or most preferred features do not work as expected	XII. Many required features work correctly	XII. All required features work correctly	**3**	
Sub Total & comments					

Submission instructions

1. Write a README file for detailed notes regarding the functionality of the corresponding code, and a set of instructions on how to run them.
2. Demonstration of the solution is also required for evaluation. Each member must demo his/her work only. It is your responsibility to ensure your software works in the lab PC and it is ready for demo without bugs/errors. Other instructions will be given later.
3. Completely fill Mark Allocation Sheet and submit it with your assignment. Failing to do so may result in a deduction of 50% marks.
4. This assignment can be submitted in groups of up to 3 members. Assign a group leader and submit the assignment through the group leader's moodle account. You have to submit just one zipped file of your project. The submission filename should read A1_Sxxx_Syyy_Szzz.zip or A1_Sxxx_Syyy_Szzz.rar where Sxxx, Syyy and Szzz are student ids of the group members. For example A1_S11003232_S01004488.zip or A1_S11003232_S01004488_S11015566.zip. **Incorrect/late submission will result in a high penalty.**
5. Marks are allocated for standard programming, your creativity, ease of use and demonstrating an error-free application.

Mark Allocation Sheet

After having discussed as group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group Name Potential

Project manager Ronil Prasad

Member ID	Percentage contribution of allocated task
S11231541	33%
S11229535	33%
S11231502	33%

Certification		
ID	Member name	Signature
S11231541	Ronil Prasad	signature in the folder
S11229535	Praheel Kumar	
S11231541	Shivan Prasad	

Due Date – please refer to Moodle.

Assessments mapping with CBOK

Core Body of Knowledge		CS214	Assign1	Assign2	Assign3	Test 1	Test 2
ICT Professional Knowledge	Ethics						
	Professional expectations						
	Teamwork concepts/issues						
	Communication	M					
	Societal Issues/Legal issues/Privacy						
	Understanding the ICT profession						
ICT Problem Solving:	Abstraction	M	✓				
	Design	M	✓				
Technology Resources	Hardware and Software Fundamentals						
	Data and Information Management	M	✓				
	Networking						
Technology Building	Human Factors						
	Programming	M	✓				
	Systems Development / Acquisition	M	✓				
ICT Management	IT Governance and organisational issues						
	IT Project management						
	Service management						
	Security management						